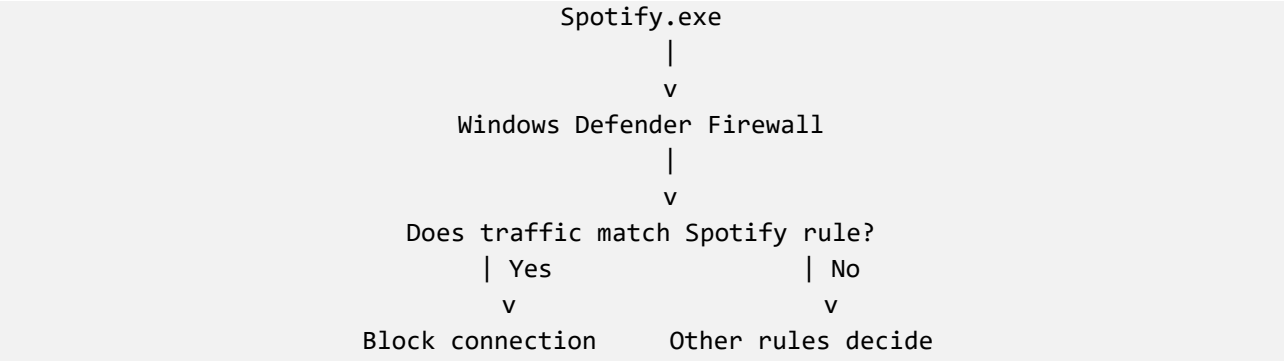
DAY 4 FIREWALL BASICS THEORY

Task 1: Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Steps

1. Press Windows + R, type wf.msc, and press Enter to open Windows Defender Firewall with Advanced Security.
2. Select Outbound Rules, then click New Rule.
3. Select Program and click Next.
4. Select This program path and browse to the actual Spotify.exe file on the computer.
5. Select Block the connection and click Next.
6. Select Domain, Private, and Public profiles if the block must apply on all network types.
7. Name the rule Block Spotify - Outbound and click Finish.
8. Repeat the same steps under Inbound Rules and name the second rule Block Spotify - Inbound.
9. Open Spotify and test whether its network features are blocked. Ensure both rules are enabled.

Diagram



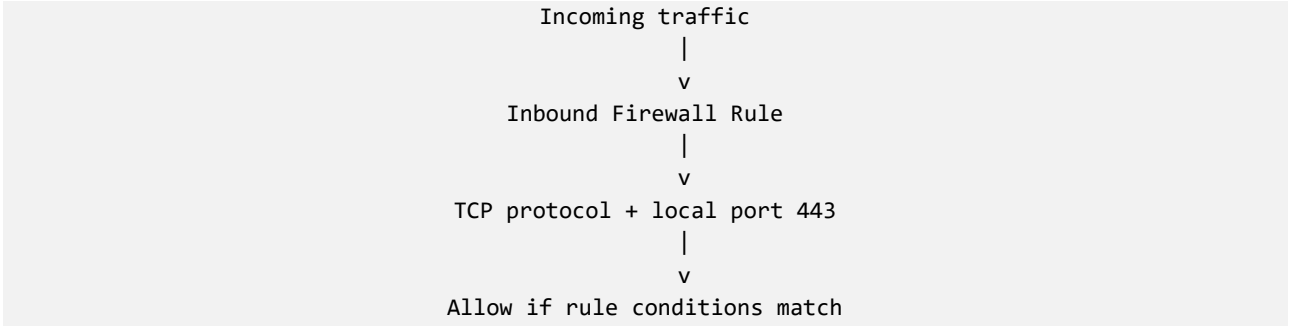
DAY 4 FIREWALL BASICS THEORY

Task 2: Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Steps

1. Press Windows + R, type wf.msc, and press Enter.
2. Select Inbound Rules and click New Rule.
3. Choose Port and click Next.
4. Select TCP and enter 443 in Specific local ports. Click Next.
5. Choose Allow the connection and click Next.
6. Select the required network profiles. Use only the profiles needed for the lab.
7. Name the rule Allow Chrome Inbound TCP 443 and click Finish.
8. To make the rule program-specific, create or configure a Program rule using the actual Chrome.exe path and appropriate protocol/port settings where supported.

Diagram



DAY 4 FIREWALL BASICS THEORY

Task 3: Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Example

Sample remote IP address: 123.45.67.89. This is an example value; use an authorized IP address for a real lab.

Steps to block traffic from the IP address

1. Open Windows Defender Firewall with Advanced Security by running wf.msc.

2. Select Inbound Rules and click New Rule. Choose Custom, or create a suitable Program/Port rule.

3. Configure the program, protocol, and ports if required, then continue to the Action page.

4. Select Block the connection. Choose the required profiles and give the rule a name, for example Block inbound from sample IP.

5. After creating the rule, double-click it and open the Scope tab.

6. Under Remote IP address, select These IP addresses and click Add.

7. Enter 123.45.67.89, confirm, then click Apply and OK.

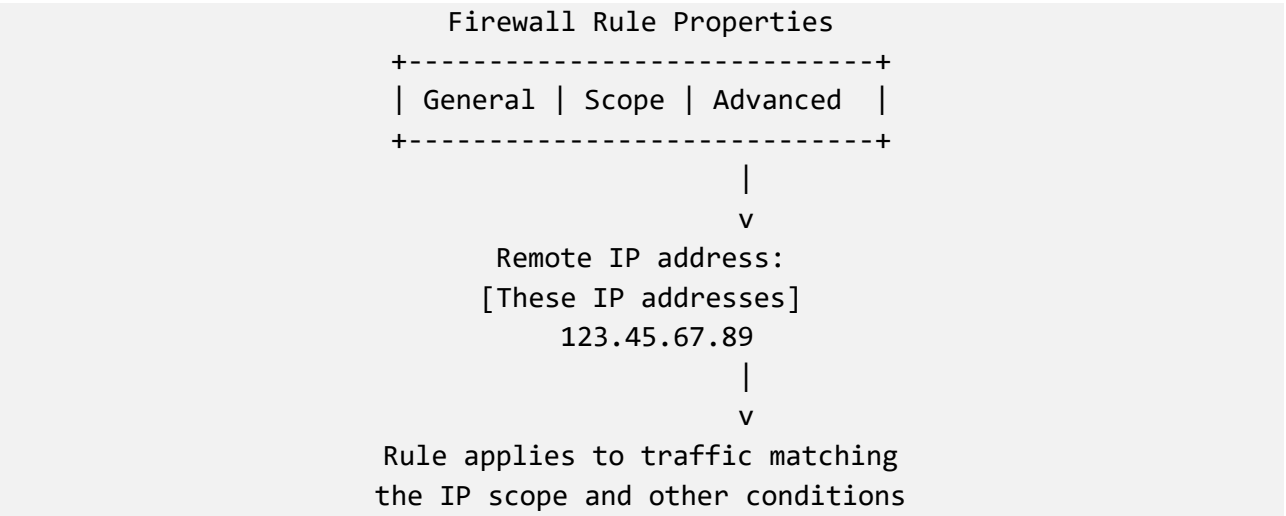
8. Ensure the rule is enabled and test it only in an authorized environment.

To allow traffic instead

Create a similar rule, but select Allow the connection on the Action page. An allow rule may not override a matching block rule or centrally managed security policy.

DAY 4 FIREWALL BASICS THEORY

Diagram



Verification Commands / Checks

Windows Defender Firewall uses a graphical interface for these steps. Reopen the rule Properties → Scope and verify that the remote IP is listed and the rule is enabled.

DAY 4 FIREWALL BASICS THEORY

Task 4: Compare hardware firewalls and software firewalls by listing three key differences and a use case for each.

Answer

Difference	Hardware Firewall	Software Firewall
Location	Usually a router or dedicated appliance at the network boundary.	Installed on an individual computer.
Coverage	Can protect many devices on a network.	Primarily protects the device where it runs.
Rule control	Often focuses on network-wide IP, port, and service rules.	Can often create application-specific rules for programs.
Management	Usually configured centrally by a network administrator.	Configured locally or by endpoint management tools.

Use Cases

Hardware firewall: protect multiple computers in a school or office network. Software firewall: block a specific application or restrict traffic on an individual computer. Using both provides layered protection.